

1. HEADER

Name: N.SATHWIK

Roll No.: 192525077

Course Code & CO Reference: [CSA0926], [PROGRAMMING IN JAVA]

Assignment Set: Direct-Descriptive

Question Number: Q23

GitHub Repository Link: [<https://github.com/sathwiknagella/CSA0926-JAVAPROGRAMMING/upload/main>]

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ASSIGNED QUESTION

Q23. ITERATING A HASHMAP WITH ENTRYSET()

Write a Java program that creates a HashMap containing several key-value pairs, and then iterates over all of its entries using the `entrySet()` method combined with a loop over `Map.Entry` objects. For each entry, display the key and its corresponding value in a clearly formatted line.

CONCEPTS: HashMap, `entrySet()`, `Map.Entry`.

2. PROBLEM UNDERSTANDING

The program creates a HashMap to store several student names and their corresponding marks. It uses the `entrySet()` method and a for-each loop with `Map.Entry` objects to access and display every key-value pair clearly.

3. APPROACH / DESIGN NOTES

- Used the HashMap class to store student names and marks as key-value pairs.
- Used String as the key type and Integer as the value type.
- Added the key-value pairs using the `put()` method.
- Used the `entrySet()` method to access all entries in the HashMap.
- Used a for-each loop with `Map.Entry<String, Integer>` to retrieve each key and its corresponding value.
- Used `getKey()` and `getValue()` methods to display the student name and marks.
- `entrySet()` was chosen because it allows both the key and value to be accessed together.

4. SOURCE CODE

```
import java.util.HashMap;

import java.util.Map;


public class Main
{
    public static void main(String[] args)
    {
        HashMap<String, Integer> marks = new HashMap<>();

        marks.put("Sathwik", 90);
        marks.put("Ravi", 85);
        marks.put("Kiran", 78);
        marks.put("Arun", 92);

        System.out.println("Student Marks:");

        for (Map.Entry<String, Integer> entry : marks.entrySet())
        {
            System.out.println(
                "Name: " + entry.getKey()
                + " | Marks: " + entry.getValue()
            );
        }
    }
}
```

5. TEST CASES

Test Case No.	Input	Expected Output	Actual Output
1	HashMap containing four student names and marks	All four student names and their corresponding marks are displayed	All four student names and their corresponding marks are displayed
2	HashMap containing one student name and mark	The single student name and corresponding mark are displayed	The single student name and corresponding mark are displayed
3	Empty HashMap	Only the heading Student Marks: is displayed	Only the heading Student Marks: is displayed

NOTE: HashMap does not maintain insertion order. Therefore, the entries may appear in a different order when the program is executed.

6. SCREENSHOTS OF EXECUTION

TEST CASE 1: MULTIPLE KEY-VALUE PAIRS

INPUT:

```
marks.put("Sathwik", 90);
```

```
marks.put("Ravi", 85);
```

```
marks.put("Kiran", 78);
```

```
marks.put("Arun", 92);
```

OUTPUT:

Student Marks:

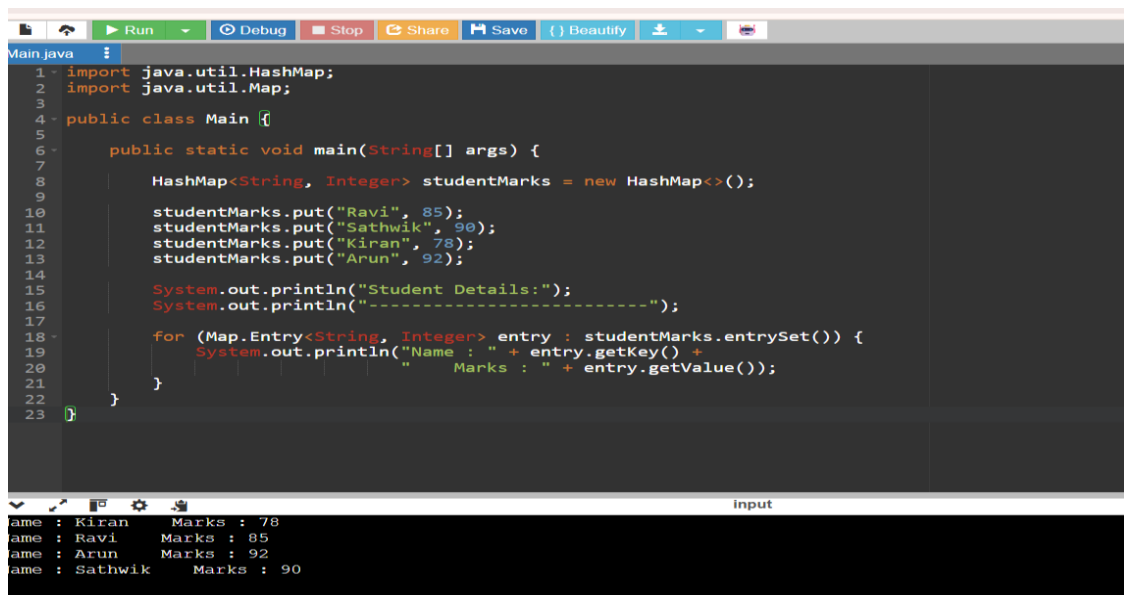
Name: Ravi | Marks: 85

Name: Arun | Marks: 92

Name: Sathwik | Marks: 90

Name: Kiran | Marks: 78

Insert Screenshot 1 here



```
1 import java.util.HashMap;
2 import java.util.Map;
3
4 public class Main {
5
6     public static void main(String[] args) {
7
8         HashMap<String, Integer> studentMarks = new HashMap<>();
9
10        studentMarks.put("Ravi", 85);
11        studentMarks.put("Sathwik", 90);
12        studentMarks.put("Kiran", 78);
13        studentMarks.put("Arun", 92);
14
15        System.out.println("Student Details:");
16        System.out.println("-----");
17
18        for (Map.Entry<String, Integer> entry : studentMarks.entrySet()) {
19            System.out.println("Name : " + entry.getKey() +
20                               " Marks : " + entry.getValue());
21        }
22    }
23 }
```

name : Kiran Marks : 78
name : Ravi Marks : 85
name : Arun Marks : 92
name : Sathwik Marks : 90

TEST CASE 2: SINGLE KEY-VALUE PAIR

INPUT:

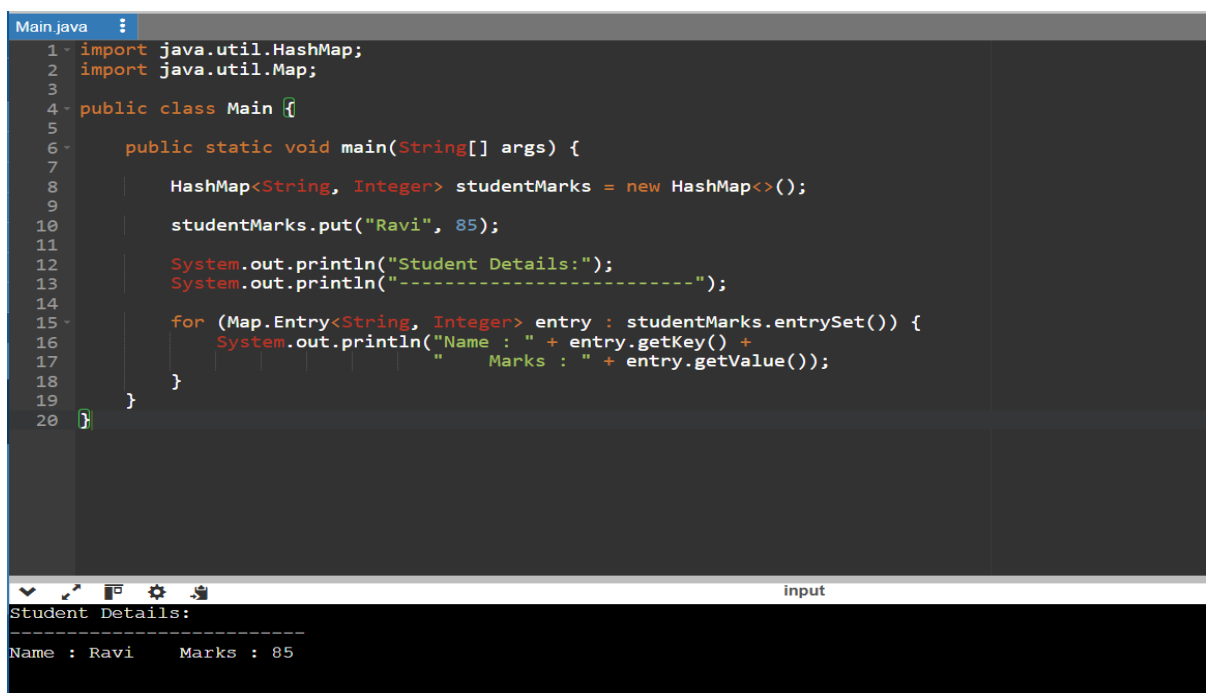
marks.put("Sathwik", 90);

OUTPUT:

Student Marks:

Name: Sathwik | Marks: 90

Insert Screenshot 2 here



```
1 import java.util.HashMap;
2 import java.util.Map;
3
4 public class Main {
5
6     public static void main(String[] args) {
7
8         HashMap<String, Integer> studentMarks = new HashMap<>();
9
10        studentMarks.put("Ravi", 85);
11
12        System.out.println("Student Details:");
13        System.out.println("-----");
14
15        for (Map.Entry<String, Integer> entry : studentMarks.entrySet()) {
16            System.out.println("Name : " + entry.getKey() +
17                               " Marks : " + entry.getValue());
18        }
19    }
20 }
```

Student Details:

Name : Ravi Marks : 85

TEST CASE 3: EMPTY HASHMAP

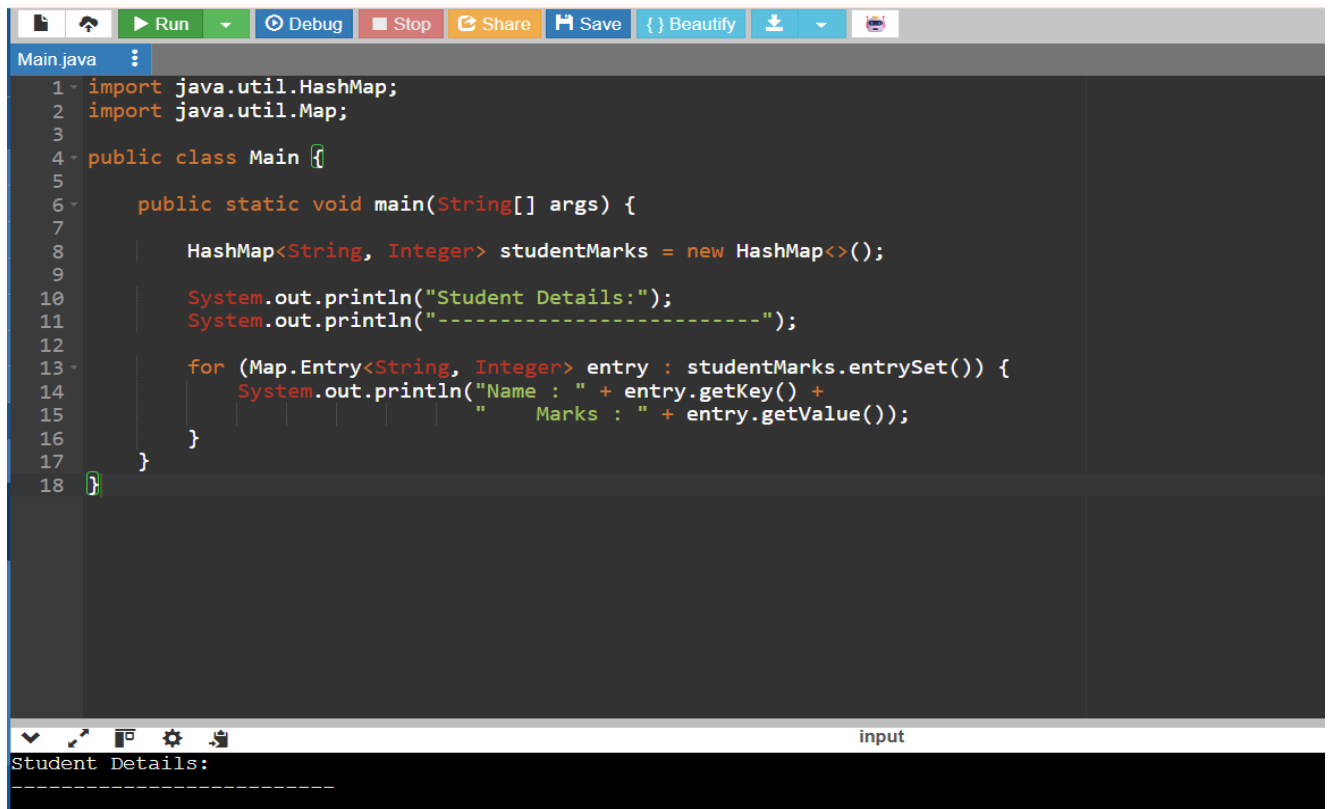
INPUT:

// No key-value pairs are added

OUTPUT:

Student Marks:

Insert Screenshot 3 here



```
1 import java.util.HashMap;
2 import java.util.Map;
3
4 public class Main {
5
6     public static void main(String[] args) {
7
8         HashMap<String, Integer> studentMarks = new HashMap<>();
9
10        System.out.println("Student Details:");
11        System.out.println("-----");
12
13        for (Map.Entry<String, Integer> entry : studentMarks.entrySet()) {
14            System.out.println("Name : " + entry.getKey() +
15                               "      Marks : " + entry.getValue());
16        }
17    }
18 }
```

input

Student Details:

7. REFLECTION

I learned how to store and manage data using a HashMap. I also learned how the `entrySet()` method and `Map.Entry` can be used together to access and display each key with its corresponding value.

8. DECLARATION

“This is my own work. Any AI/external tool assistance used is disclosed above.”

AI/external tool assistance used: AI assistance was used to understand the assignment format and explain the Java concepts. The program was reviewed, executed, and verified by me.

